

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Quantum Computing Fundamentals for Beginners

Course Code: C882

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	2 October 2026	Initial release of the Quantum Computing Fundamentals for Beginners lesson plan: two 9:30 am – 5:30 pm days (15 instructional hours) across the four topics, mapped to the 17 hands-on IBM Quantum Composer labs and the v1.0 95-slide deck.	Dr Alfred Ang

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Course Information

Course Title	Quantum Computing Fundamentals for Beginners
Course Code	C882
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 7.5 instructional hours per day (15 hours)
Daily Timing	9:30 am – 5:30 pm on both days (7.5 instructional hours each). The single 30-minute lunch break is excluded from instructional time; refreshments are taken during activity time.
Mode	Instructor-led, hands-on quantum circuit labs on IBM Quantum Composer (browser-based simulator and real quantum hardware)
Target Audience	Beginners with no prior quantum experience; basic computer literacy and a positive learning attitude
Trainer	Dr Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Identify and use appropriate quantum computing platforms for testing quantum gates and algorithms
- LO2: Utilise and conduct quantum gates and logic to derive solutions
- LO3: Perform coding and conduct testing of quantum algorithms
- LO4: Diagnose outcomes, propose, and implement changes for quantum computing applications

Course Outline

Topic	Sub-topics	Hands-on labs
Topic 01: Introduction to Quantum Computing	Key Principles of Quantum Computing; Quantum Computing Platforms	Labs 1–3
Topic 02: Quantum States and Gates	Quantum State Vector Representations; Quantum Circuits and Quantum Gates	Labs 4–7
Topic 03: Quantum Algorithms	Coding Quantum Algorithms; Testing Quantum Algorithms	Labs 8–13
Topic 04: Quantum Computing Applications	Introduction to Quantum Error Correction; Introduction to Quantum Machine Learning	Labs 14–17

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Course Schedule

Each training day runs 9:30 am – 5:30 pm and comprises 7.5 instructional hours. The single 30-minute lunch break is excluded from instructional time.

Day 1 — Quantum Principles, States and Gates

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, trainer and learner introductions, ground rules, course outline and learning outcomes (Slides from p.1)
10:00–11:00	60 min	**Topic 1 — Introduction to Quantum Computing: what a quantum computer is, the qubit, the Bloch sphere, physical qubit implementations, superposition, interference and entanglement (concepts + demonstration) (Slides from p.9)**
11:00–12:15	75 min	Topic 1 continued — separability, the five postulates of quantum mechanics, quantum computing platforms and industry use cases. Hands-on: Lab 1: Explore the Qubit on the Bloch Sphere; Lab 2: Observe Superposition and Interference
12:15–12:45	30 min	Lunch break
12:45–13:15	30 min	Hands-on: Lab 3: Create Your First Entangled Pair (Bell State)
13:15–14:45	90 min	**Topic 2 — Quantum States and Gates: state vectors and Dirac notation, unitary gates, the Pauli gates X/Y/Z, the Hadamard gate, rotation and phase gates (concepts + demonstration) (Slides from p.28)**
14:45–16:15	90 min	Topic 2 continued — CNOT, CZ, SWAP and Toffoli; Bell and GHZ states; phase kickback. Hands-on: Lab 4: Single-Qubit Gates — Pauli X, Y, Z and Hadamard; Lab 5: Rotation and Phase Gates — RX, RY, RZ, S and T
16:15–17:15	60 min	Hands-on: Lab 6: Two-Qubit Gates — CNOT, CZ, SWAP and Toffoli; Lab 7: Bell States, GHZ States and Phase Kickback
17:15–17:30	15 min	Day 1 recap, Q&A and close

Total instructional time: 450 minutes (7.5 hours).

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Day 2 — Quantum Algorithms and Applications

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap
9:45–11:00	75 min	**Topic 3 — Quantum Algorithms: the no-cloning theorem, quantum teleportation, superdense coding and BB84 quantum key distribution (concepts + demonstration) (Slides from p.46)**
11:00–12:20	80 min	Hands-on: Lab 8: Quantum Teleportation; Lab 9: Superdense Coding; Lab 10: BB84 Quantum Key Distribution
12:20–12:50	30 min	Lunch break
12:50–13:35	45 min	**Topic 3 continued — the query/oracle model, phase kickback in practice, Deutsch and Deutsch-Jozsa, Bernstein-Vazirani, Grover's search and the Quantum Fourier Transform (concepts + demonstration)**
13:35–14:55	80 min	Hands-on: Lab 11: The Deutsch and Deutsch-Jozsa Algorithms; Lab 12: The Bernstein-Vazirani Algorithm; Lab 13: Grover's Search Algorithm
14:55–15:40	45 min	**Topic 4 — Quantum Computing Applications: quantum error correction (bit-flip, phase-flip, Shor, Steane and surface codes), fault tolerance, simulator versus real hardware, and quantum machine learning (concepts + demonstration) (Slides from p.70)**
15:40–17:00	80 min	Hands-on: Lab 14: Quantum Error Correction — The Bit-Flip Code; Lab 15: Quantum Error Correction — The Phase-Flip Code; Lab 16: Simulator vs Real Quantum Hardware; Lab 17: Introduction to Quantum Machine Learning
17:00–17:30	30 min	Course summary, Q&A and course feedback (Slides from p.87)

Total instructional time: 450 minutes (7.5 hours).

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Lab Reference (aligned to the four topics)

Topic	Learning outcome	Labs	Platform
Topic 01: Introduction to Quantum Computing	LO1	Lab 1, Lab 2, Lab 3	IBM Quantum Composer
Topic 02: Quantum States and Gates	LO2	Lab 4, Lab 5, Lab 6, Lab 7	IBM Quantum Composer
Topic 03: Quantum Algorithms	LO3	Lab 8, Lab 9, Lab 10, Lab 11, Lab 12, Lab 13	IBM Quantum Composer
Topic 04: Quantum Computing Applications	LO4	Lab 14, Lab 15, Lab 16, Lab 17	IBM Quantum Composer

Detailed Lab Schedule

Lab	Title	Learning Outcome	Tool
Lab 1	Explore the Qubit on the Bloch Sphere	LO1 — identify and use appropriate quantum computing platforms	Bloch Sphere Simulator (bloch.kherb.io)
Lab 2	Observe Superposition and Interference	LO1 — identify key principles of quantum computing	IBM Quantum Composer
Lab 3	Create Your First Entangled Pair (Bell State)	LO1/LO2 — use a quantum platform to demonstrate entanglement	IBM Quantum Composer
Lab 4	Single-Qubit Gates — Pauli X, Y, Z and Hadamard	LO2 — utilise quantum gates and logic to derive solutions	IBM Quantum Composer
Lab 5	Rotation and Phase Gates — RX, RY, RZ, S and T	LO2 — utilise quantum gates and logic to derive solutions	IBM Quantum Composer
Lab 6	Two-Qubit Gates — CNOT, CZ, SWAP and Toffoli	LO2 — utilise quantum gates and logic to derive solutions	IBM Quantum Composer
Lab 7	Bell States, GHZ States and Phase Kickback	LO2 — utilise quantum gates and logic to derive solutions	IBM Quantum Composer
Lab 8	Quantum Teleportation	LO3 — perform coding and conduct testing of quantum algorithms	IBM Quantum Composer
Lab 9	Superdense Coding	LO3 — perform coding and conduct testing of quantum algorithms	IBM Quantum Composer
Lab 10	BB84 Quantum Key Distribution	LO3 — perform coding and conduct testing of quantum algorithms	IBM Quantum Composer
Lab 11	The Deutsch and Deutsch-Jozsa Algorithms	LO3 — perform coding and conduct testing of quantum algorithms	IBM Quantum Composer
Lab 12	The Bernstein-Vazirani Algorithm	LO3 — perform coding and conduct testing of quantum algorithms	IBM Quantum Composer
Lab 13	Grover's Search Algorithm	LO3 — perform coding and conduct testing of quantum algorithms	IBM Quantum Composer

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Lab 14	Quantum Error Correction — The Bit-Flip Code	LO4 — diagnose outcomes and implement changes	IBM Quantum Composer
Lab 15	Quantum Error Correction — The Phase-Flip Code	LO4 — diagnose outcomes and implement changes	IBM Quantum Composer
Lab 16	Simulator vs Real Quantum Hardware	LO4 — diagnose unintended outcomes and propose changes	IBM Quantum Composer, IBM Quantum Platform
Lab 17	Introduction to Quantum Machine Learning	LO4 — propose and implement changes for quantum computing applications	IBM Quantum Composer

Training Resources

- Course slides: Quantum Computing Fundamentals for Beginners (C882)-v1.0.pptx (Trainer) and the learner PDF.
- Learner Guide: LG-Quantum Computing Fundamentals for Beginners (C882).docx — contains the full step-by-step instructions for all 17 labs.
- Lab files: labs/lab-01 … lab-17 with the OpenQASM source for every circuit.
- Platform: IBM Quantum Composer — https://quantum.cloud.ibm.com/composer (free IBM Quantum account required; browser only).
- LMS/TMS portal: https://lms-tms.tertiaryinfotech.com for the course materials.
- Each learner requires a laptop with a modern browser and internet access.

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